

2MARKET DATA ANALYSIS REPORT



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Total word count without Front page, Table of contents and Appendix:

1,099

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1. Background/context of the report

1.1 Establish the need

S.N	2Market needs	Objectives of the report
1.	Understand the customer demography.	Provide insights into the composition of the customer demography by country, age, Income, and marital status.
2.	Understand the best-selling products and whether the best-selling product varies by customer demography.	Provide insights into the best-selling products by customer demography.
3.	Understand the most effective advertising channels.	Provide insights on choosing the most effective advertising channels when communicating about their products to customers from different demography.

1.2 Justify the need

By fulfilling the objectives of this report, 2Market will have an understanding of their customer demography, their best-selling products and the most and least effective advertising channels.

1.3 Contextualize the problem

This problem is faced by the marketing team at 2Market.

1.4 Write the problem statement

The problem is,

“2market needs to create a marketing campaign and for that, 2 market needs to understand their customers, products and advertising channels”.

2. Analytical Approach

I have cleaned the marketing dataset in Excel.

I have used Trim function on all the column titles to remove blank space. Some of the column titles were not clear. I have used the information in the Metadata to re-title those columns. I have renamed the country abbreviations for Spain from “SP” to “ESP, and for South Africa from “SA” to “RSA”.

Using “TYPE” function, I found excel recognized the values in Column E (Income) and Column H (Dt_Customer) as text and not number. After web research, I found that removing the “\$” sign present in front of the number in Column E (Income) would permit Excel to recognize the values in the Column E as a number. I inserted ~\$ in “Find” and didn’t insert anything in “Replace”. In this way, finally the values in Column E is recognised as number by Excel.

After using Text to Column function, Excel still didn’t recognize the “Dt_Customer” column as having date values. I decided to copy and paste the values of only the “Dt_Customer” column into a new worksheet. Then, I used Text and Column function to split the dates. Then, in a new

column, I used Date function to specify day, month, year. Excel generated date into the specified cell. Then, I dragged the formula to all the rows and then excel generated this:

Dt_Customer	Month	Day	Year	
6/16/14	6	16	14	16/06/1914
6/15/14	6	15	14	15/06/1914
5/13/14	5	13	14	13/05/1914
05/11/2014	5	11	2014	11/05/2014
04/08/2014	4	8	2014	08/04/2014
3/17/14	3	17	2014	17/03/2014
1/29/14	1	29	2014	29/01/2014
1/18/14	1	18	14	18/01/1914
01/11/2014	1	11	2014	11/01/2014
01/11/2014	1	11	2014	11/01/2014
12/27/13	12	27	13	27/12/1913
12/09/2013	12	9	2013	09/12/2013
12/07/2013	12	7	2013	07/12/2013
10/16/13	10	16	13	16/10/1913
10/05/2013	10	5	2013	05/10/2013
09/11/2013	9	11	2013	11/09/2013
09/11/2013	9	11	2013	11/09/2013
08/01/2013	8	1	2013	01/08/2013
7/23/13	7	23	13	23/07/1913

Excel recognized all the years that were written in “/12”, “/13” and “/14” format as 1912, 1913 and 1914. Through web research, I found a formula. Using the formula, the “year” has been changed from “1912”, “1913” and “1914” into “2012”, “2013”, and “2014. Below is a screen shot of the formula.

Get & Transform Data

Queries & Connections

Sort &

✕

✓

<

I have removed three records from the marketing dataset: ID “11004”, “1150”, and “7829” as these records had higher age value than the oldest person alive¹. I have used 2024 as a base year to calculate the age of customers.

1

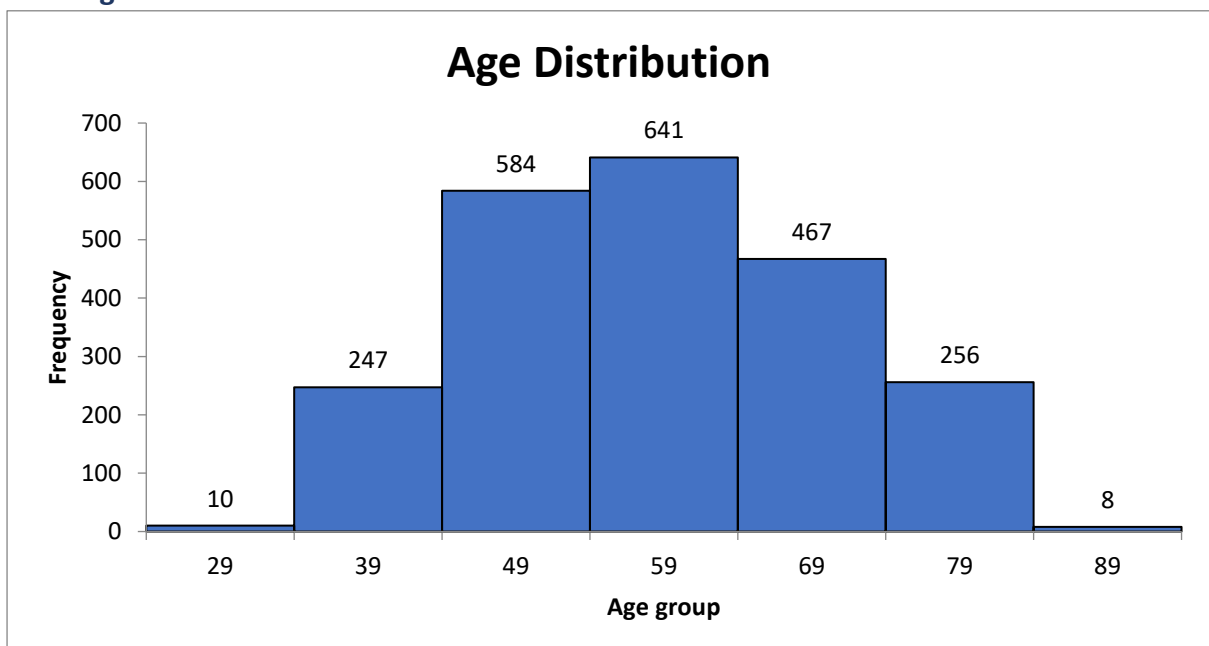
https://www.google.com/search?q=oldest+human+alive&og=oldest+human+alive&gs_lcrp=EgZjaHJvbWUqCQgAEEUYOxiABDIJCAAQRrg7GIAEMgclARAAGIAEMgclAhAAGIAEMgclAxAAGIAEMgclBBAAGIAEMgclBRAAGIAEMgOIbHAAGIYDGAEGloFMgOIbXAAGIYDGAEGloFMgOICBAAGIYDGAEGloFMgOICRAAGIYDGAEGloF0gEIMjcwN2owajeoAgCwAgA&sourceid=chrome&ie=UTF-8

2.	Advertising channels effectiveness	The rationale for designing this dashboard is to achieve the final objective of this report; 1. Provide insights on choosing the most effective advertising channels when communicating about their products to customers from different customer demography. I have used a bar chart to convey information about the effectiveness of different advertising channels by customer demography. To interact with the dashboard, three filters are provided; country, age group and marital status.
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4. Patterns, trends and insights

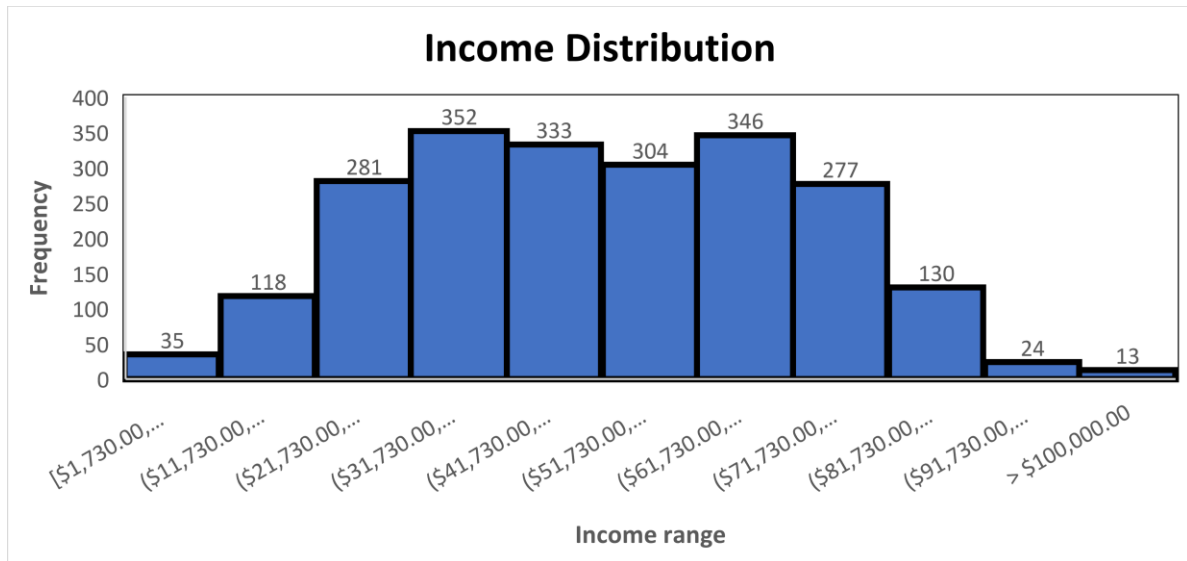
4.1 Customer Demography

4.1.1 Age Distribution



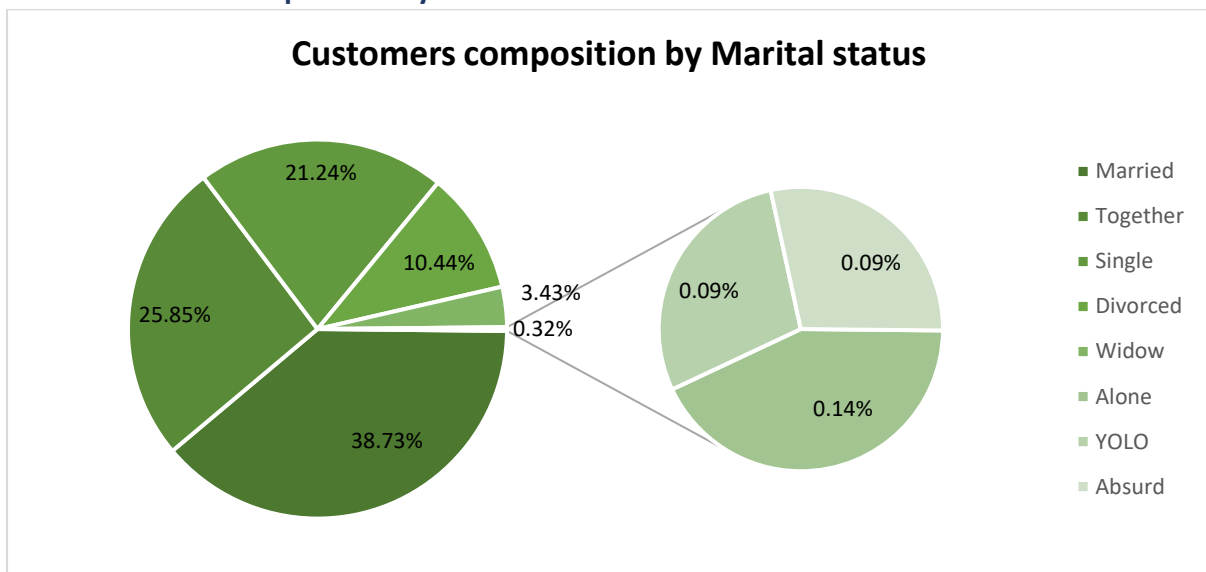
The age distribution has a symmetric distribution with most values placed in the centre. The average age of the customer is 54 years old.

4.1.2 Income Distribution



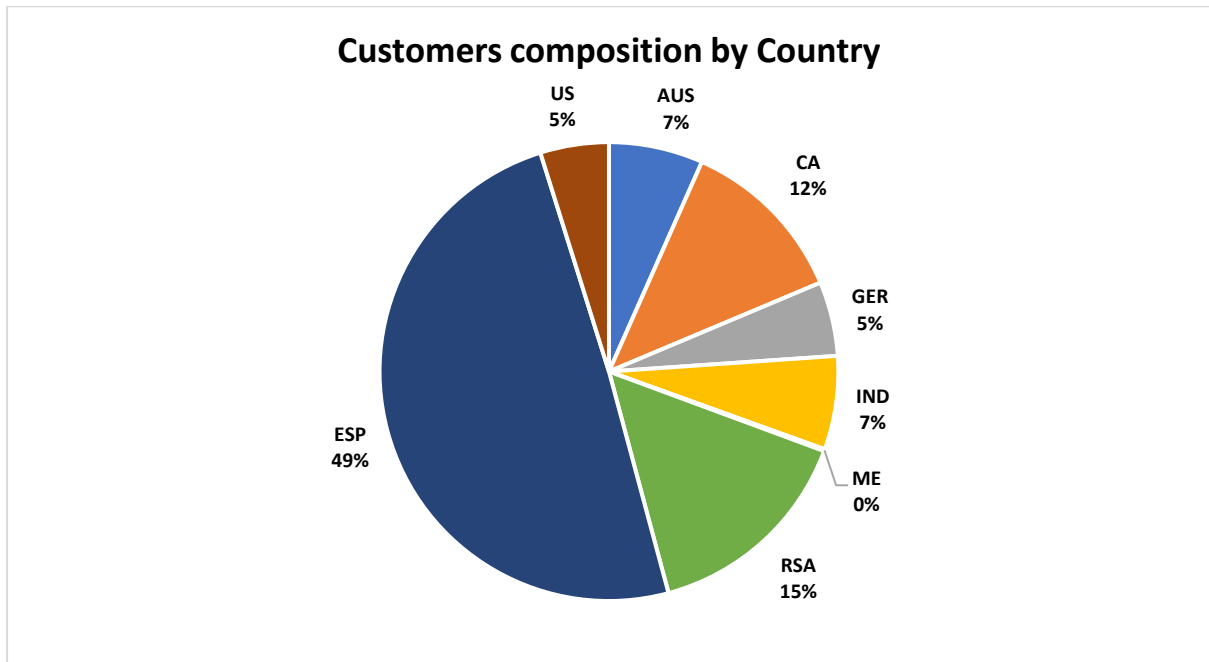
Most data values is concentrated between \$21,730-\$61,729 annual income range. The average annual income of customers is \$52,237.

4.1.3 Customers composition by Marital status



Married customers are the largest customer group by marital status.

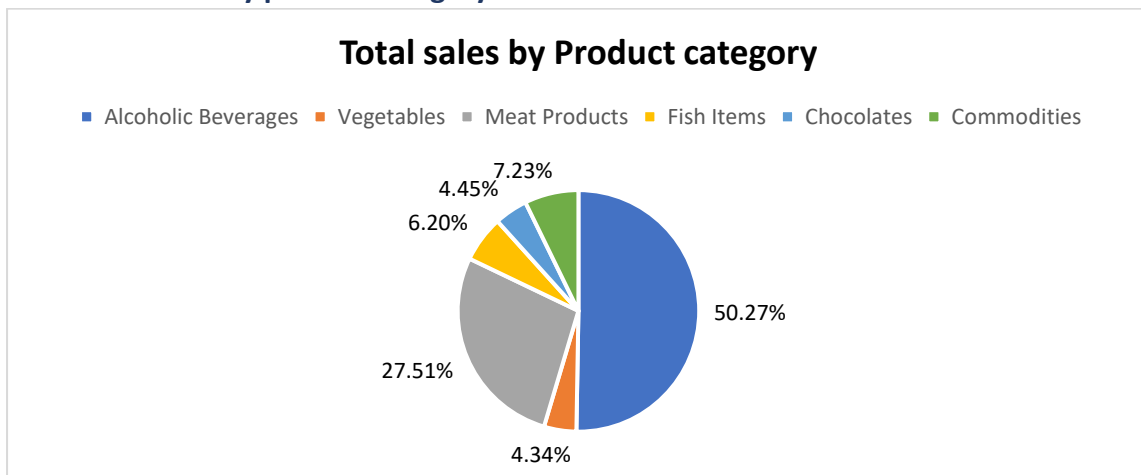
4.1.4 Customers composition by Country



Spain have the largest customer share.

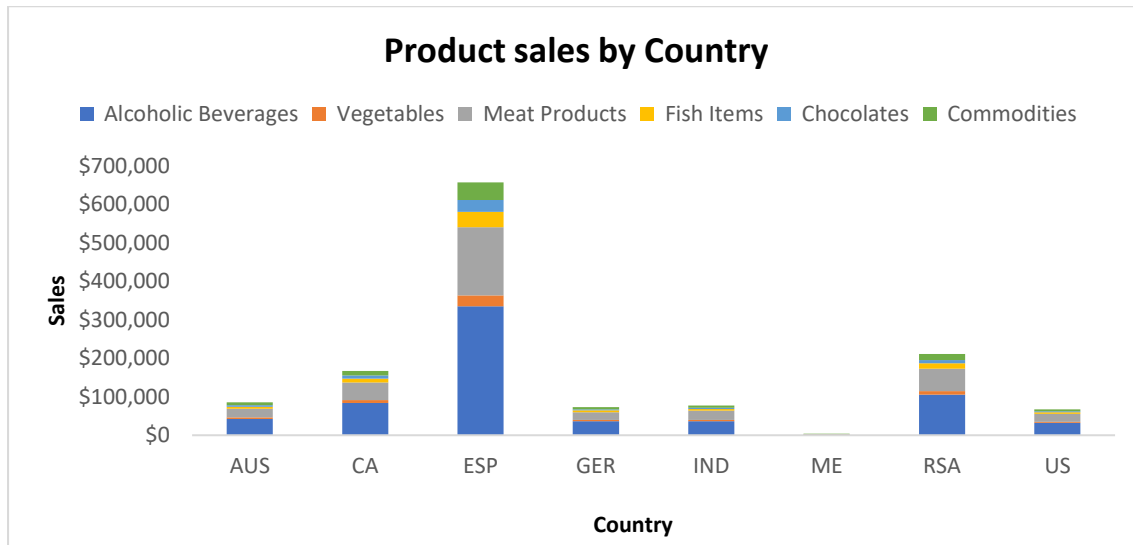
4.2 Best-selling products overall and by Customer demography

4.2.1 Total sales by product category



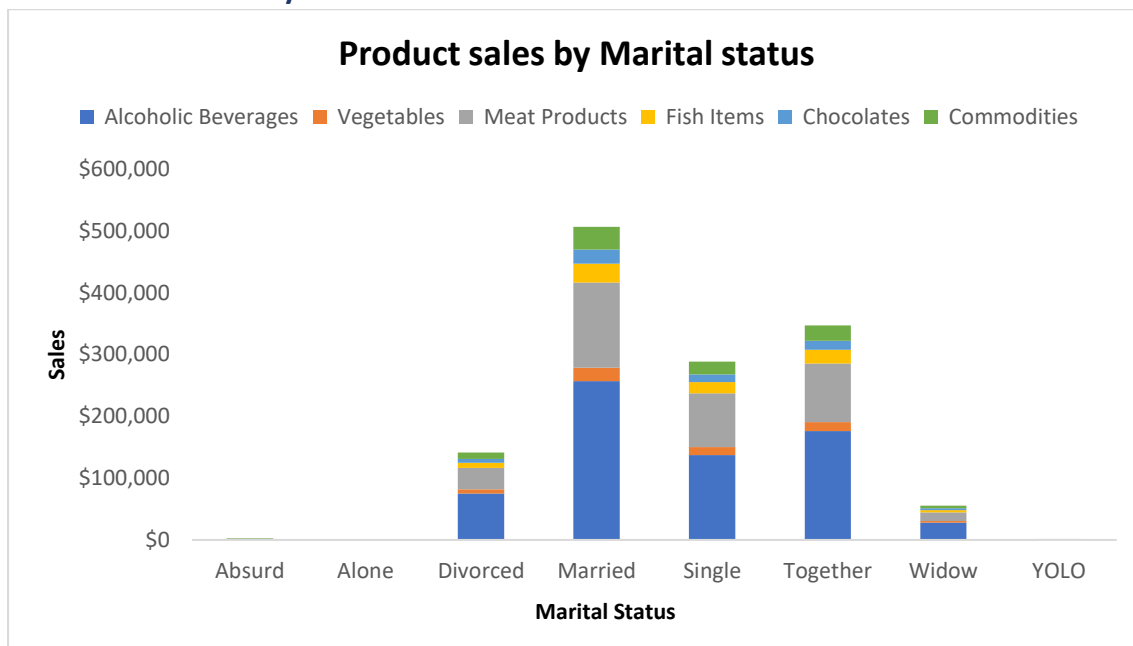
Alcoholic beverages is the best-selling product.

4.2.2 Product sales by Country



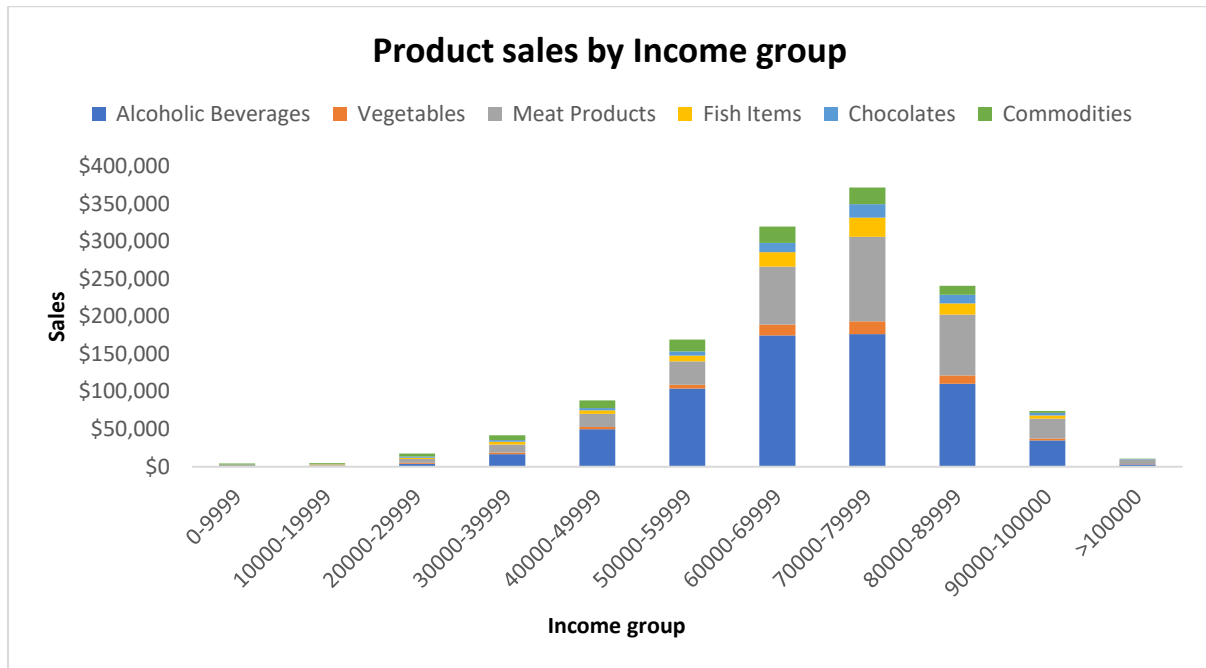
Spain have the largest proportion of sales by country. Alcoholic beverages is the best-selling product in all countries.

4.2.3 Product sales by Marital status



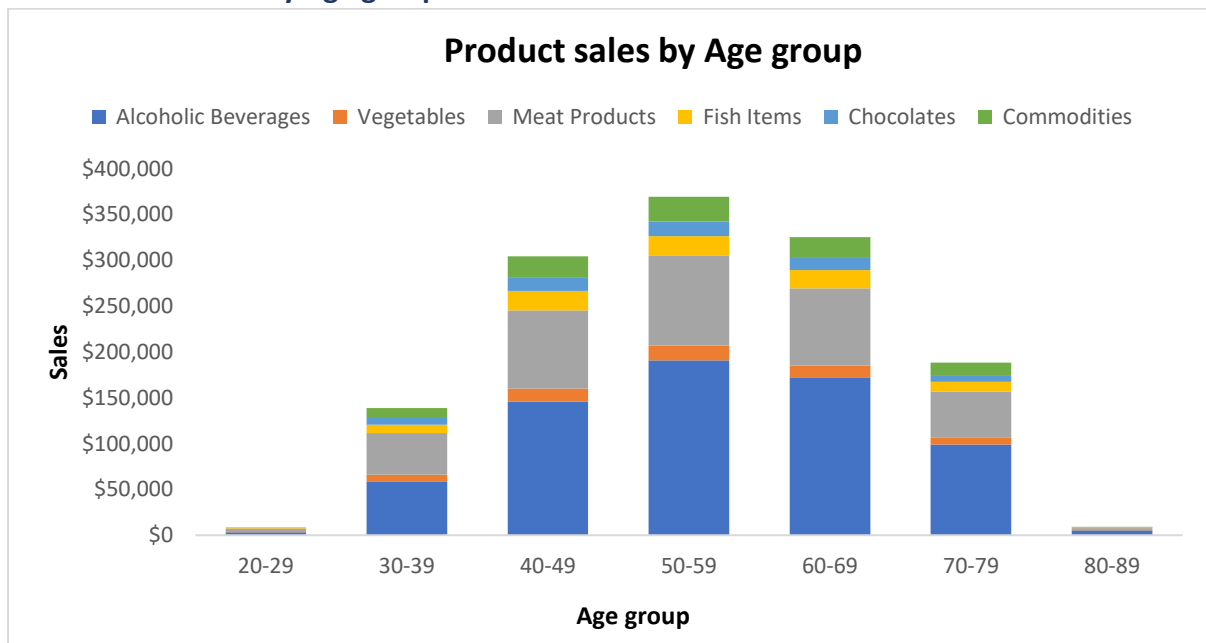
Married customers account for the largest proportion of sales. Alcoholic beverages is the best-selling product in all types of marital status.

4.2.4 Product sales by Income group



Sales generated from the income group increases as the income groups' annual income increases. Sales reaches its peak in the \$70,000-\$79,000 income group. After this, sales starts to decline. Alcoholic beverages is the best-selling product in all income groups except income group >\$100,000. Meat products is the best-selling product in the >\$100,000 income group.

4.2.5 Product sales by Age group



Most sales is generated by 50-59 years age group. The total sales increases as the age groups' age increases. The total sales reaches its peak in the 50-59 years age group. After that, the total sales decreases as the age groups' age increases. Alcohol beverages is the best-selling

product across all age groups except age group 20-29. Meat products is the best-selling product in age group 20-29.

4.3 Advertising channel effectiveness

4.3.1 Overall advertising channel effectiveness

Total_ad_conversion bigint	Twitter_ad bigint	Instagram_ad bigint	Facebook_ad bigint	Bulkmail_ad bigint	Brochure_ad bigint
660	164	161	142	163	30

The query is attached in Appendix A.

Overall, Twitter is the most effective advertising channel and the least effective advertising channel is brochure.

4.3.2 Advertising channel effectiveness by country

	country character	Twitter_ad bigint	Instagram_ad bigint	Facebook_ad bigint	Bulkmail_ad bigint	Brochure_ad bigint
1	AUS	6	12	7	9	0
2	CA	24	21	18	18	6
3	ESP	87	88	76	83	16
4	GER	11	8	7	10	2
5	IND	10	6	7	13	2
6	ME	0	0	0	1	0
7	RSA	20	21	20	21	4
8	US	6	5	7	8	0

The query is attached in appendix B.

Below, most and least effective advertising channel by country is provided.

Country	Most effective advertising channel	Least effective advertising channel
AUS	Instagram.	Brochure.
CA	Twitter.	Brochure.
ESP	Instagram.	Brochure.
GER	Twitter.	Brochure.
IND	Bulk mail.	Brochure.
ME	Bulk mail.	Every ad channel other than Bulk mail.
RSA	Instagram and Bulk mail.	Brochure.
US	Bulk mail.	Brochure.

4.3.3 Advertising channel effectiveness by Marital status

	marital_status character varying (20)	Twitter_ad bigint	Instagram_ad bigint	Facebook_ad bigint	Bulkmail_ad bigint	Brochure_ad bigint
1	YOLO	0	0	0	0	0
2	Widow	10	7	5	4	1
3	Together	42	43	32	37	12
4	Single	32	31	30	38	5
5	Married	62	66	62	63	7
6	Divorced	18	13	12	20	5
7	Alone	0	0	0	1	0
8	Absurd	0	1	1	0	0

The query is attached in appendix C.

Below, most and least effective advertising channel by country is provided.

Marital status	Most effective advertising channel	Least effective advertising channel
YOLO	NONE.	NONE.
Widow	Twitter.	Brochure.
Together	Instagram.	Brochure.
Single	Bulk mail.	Brochure.
Married	Instagram.	Brochure.
Divorced	Bulk mail.	Brochure.
Alone	Bulk mail.	Every ad channel other Bulk mail.
Absurd	Instagram and Facebook.	Every ad channel other than Instagram and Facebook.

5. Further recommendation

The effectiveness of advertising channels on age groups and income groups could be explored further. In-store sales vs online sales could be explored further.

APPENDIX A

```
SELECT
    SUM(ma."no_of_successful_lead_conversion") AS "Total_ad_conversion",
    SUM(a."twitter_ad") AS "Twitter_ad",
    SUM(a."instagram_ad") AS "Instagram_ad",
    SUM(a."facebook_ad") AS "Facebook_ad",
    SUM(a."bulkmail_ad") AS "Bulkmail_ad",
    SUM(a."brochure_ad") AS "Brochure_ad"
FROM
    public."Marketing_data" ma
LEFT JOIN
    public."ad_data" a USING ("id");
```

APPENDIX B

```
SELECT
    ma."country",
    SUM(a."twitter_ad") AS "Twitter_ad",
    SUM(a."instagram_ad") AS "Instagram_ad",
    SUM(a."facebook_ad") AS "Facebook_ad",
    SUM(a."bulkmail_ad") AS "Bulkmail_ad",
    SUM(a."brochure_ad") AS "Brochure_ad"
FROM
    public."Marketing_data" ma
LEFT JOIN
    public."ad_data" a USING ("id")
GROUP BY
    ma."country"
ORDER BY
    ma."country";
```

APPENDIX C

```
SELECT
    ma."marital_status",
    SUM(a."twitter_ad") AS "Twitter_ad",
    SUM(a."instagram_ad") AS "Instagram_ad",
    SUM(a."facebook_ad") AS "Facebook_ad",
    SUM(a."bulkmail_ad") AS "Bulkmail_ad",
    SUM(a."brochure_ad") AS "Brochure_ad"
FROM
    public."Marketing_data" ma
LEFT JOIN
    public."ad_data" a USING ("id")
GROUP BY
    ma."marital_status"
ORDER BY
    ma."marital_status" DESC;
```